

PREDICTIVE FIGHT ANALYTICS

Forecasting UFC Outcomes to Drive Engagement and Optimize Margins

UNDERSTANDING THE CHALLENGE

- The sports entertainment industry relies entirely on **anticipation**.
- Fans crave exciting finishes, and sportsbooks rely on accurate odds to protect their margins.
- Combat sports are inherently chaotic.

Predicting the unpredictable is difficult, but it is necessary to protect profits and scale the digital marketing effectively.



BUSINESS NECESSITIES

Moving from Gut Feelings to Data-Driven Accuracy

- A reliable forecasting engine.
- The ability to predict a "Finish" (KO/Submission) versus a "Decision" *before* the athletes step into the cage.
- A system that protects the risk margins while driving user engagement.



THE DATA ADVANTAGE

The Anatomy of a Fight

- **Historical Depth:** Trained models on Kaggle-verified UFC historical fight data.
- **Pre-Fight Focus:** Evaluates fighters purely on statistics known.
- **Key Metrics:** Analyzes strike volume, grappling accuracy, physical dimensions (reach/height), and historical win methods.



ENGINEERING THE DATA PIPELINE



CLEANSE

Standardized historical data, handled missing fighter records, and normalized outliers to ensure the algorithms learn from a clean, unbiased baseline.



PREVENT LEAKAGE

Strictly stripped out "hindsight" variables (like post-fight strike counts) ensuring the model only uses data available prior to the opening bell.



TARGET

Engineered a specific binary target variable: **Finish vs. Decision**. This distills complex fight dynamics into a singular, highly actionable business metric.

SOLUTIONS EXPLORED & RECOMMENDATIONS

Testing multiple approaches to find the safest, most accurate predictor for
production deployment.

HOW WE CHOSE THE FINAL MODEL



*We explored simple rules (Decision Trees) which proved too basic, and baseline models (Logistic Regression) which missed complex fighting dynamics. We recommend **Gradient Boosting** because it maximizes overall accuracy while effectively minimizing false positives—protecting us from confidently predicting a finish that doesn't happen.*

WHAT DRIVES A FIGHT FINISH?

 **Striking Output & Accuracy:** High volume strikers with historical accuracy metrics significantly increase the algorithmic probability of an early KO/TKO finish.

 **Age & Experience Differential:** Larger age gaps between competitors strongly correlate with finishes. The model heavily weights youth advantages in late-round scenarios.

 **Grappling Dominance:** Fighters defined as submission specialists with exceptionally high takedown success rates skew predictions sharply away from decisions.

RISKS & MITIGATIONS



BUSINESS RISK

Over-reliance on the model could lead to bad odds if a fighter undergoes sudden, unquantified changes (injury or camp change).

- ✓ Mitigation: Maintain "human-in-the-loop" expert oddsmakers to adjust model baselines manually.



ETHICAL RISK

The model might unintentionally bias against specific fighter demographics or regional fighting styles based on historical skew.

- ✓ Mitigation: Conduct quarterly fairness audits to ensure predictions rely strictly on performance metrics.

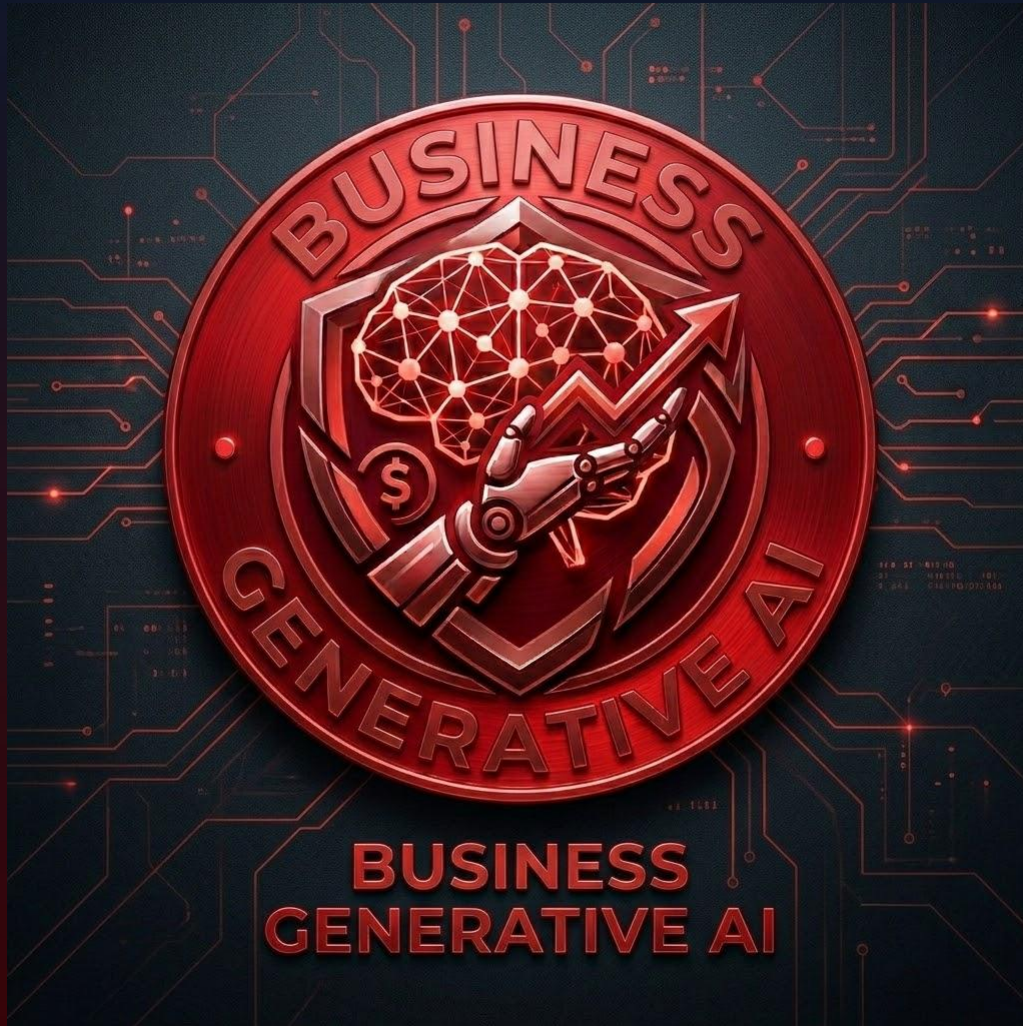


REGULATORY RISK

Gambling commissions have strict regulations regarding automated odds generation and consumer fairness.

- ✓ Mitigation: Implement strict model explainability frameworks so regulators can transparently audit how probabilities are derived.

THE "SECRET WEAPON": GENERATIVE AI



AUTOMATED FIGHT PREVIEWS

Beyond simple numerical predictions, our pipeline integrates Large Language Models (LLMs). The system digests our predictive data and automatically generates dynamic, compelling narrative summaries for upcoming fights.

This translates raw probabilities into instant, readable content ready for sports blogs, social media channels, and directly inside sports betting applications.

BUSINESS IMPACT & ROI

2x
CONTENT SCALE

SCALING FAN ENGAGEMENT

By automating pre-fight narratives and forecasting outcomes, we can double our digital content generation without adding headcount. Simultaneously, we arm the business with highly accurate prop-bet probabilities to optimize odds-making and drive new revenue streams.

PRODUCTION DEPLOYMENT ROADMAP



CLOUD HOSTING

Deploying the serialized machine learning model to scalable cloud infrastructure (AWS/GCP) to ensure high availability and robust security.



BATCH INFERENCE

Running automated predictions on a weekly batch schedule as new UFC fight cards are officially announced, optimizing compute costs.



API INTEGRATION

Creating a seamless REST API to feed predictive data directly to our internal dashboards, consumer apps, and sportsbook partners.

READY FOR PHASE 1.

Questions & Next Steps for Rollout.

IMAGE SOURCES



<https://www.kfcyumcenter.com/news/detail/record-breaking-ufc-fight-night-smashes-revenue-records-at-kfc-yum-center>

Source: [kfcyumcenter.com](https://www.kfcyumcenter.com)



<https://www.agilesherpas.com/blog/5-signs-you-need-organizational-agility>

Source: [Agilesherpas.com](https://www.agilesherpas.com)



https://www.espn.com/mma/story/_/id/31707715/ufc-fight-night-live-results-analysis-ciryl-gane-not-devastating-improving

Source: [Jeff Bottari / Zuffa LLC via Getty Images](#)

THE ANATOMY OF A **FINISH**

WHAT THE DATA ACTUALLY SAYS

3,084 FINISHES

2,705 DECISIONS

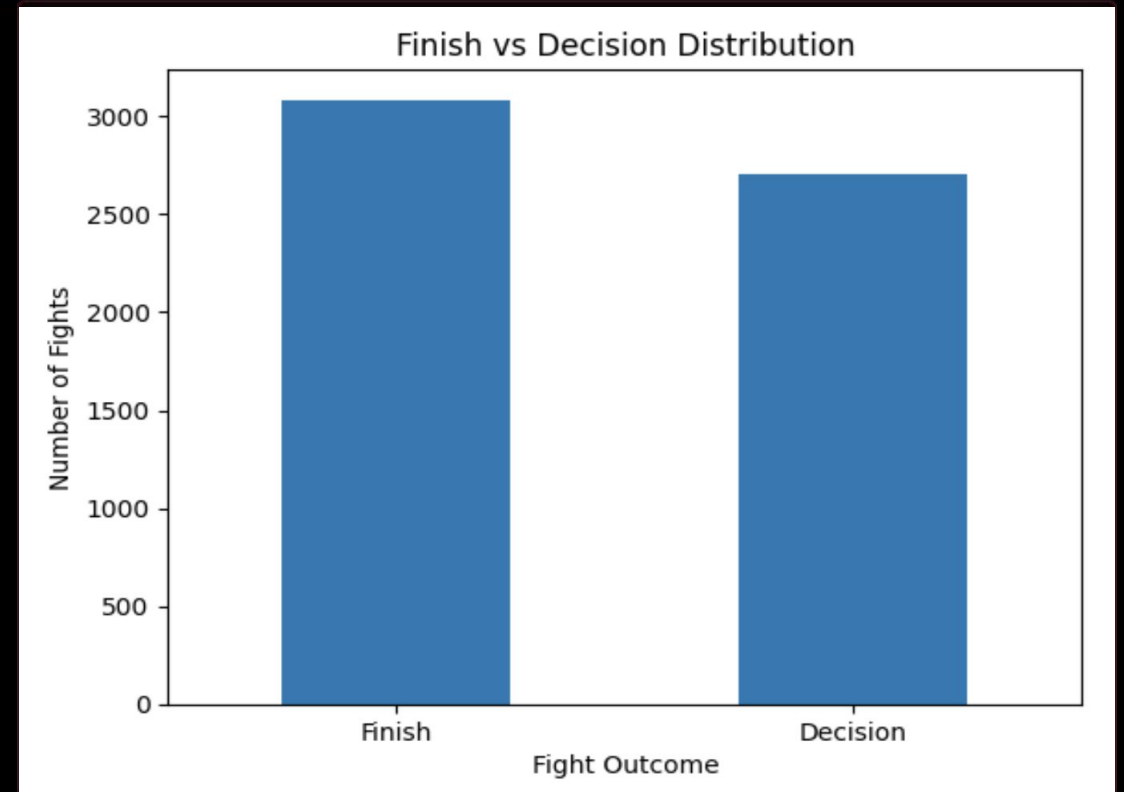
"When we watch a fight end in a spectacular knockout, it feels like pure, unpredictable chaos. Yet to a machine, it's not chaos; it's a math equation. Today, we are going to look at how we turned the blood and sweat of the Octagon into a predictive engine using over 5,700 fights."

DO STRIKERS REALLY **FINISH** MORE FIGHTS?

(EDA)

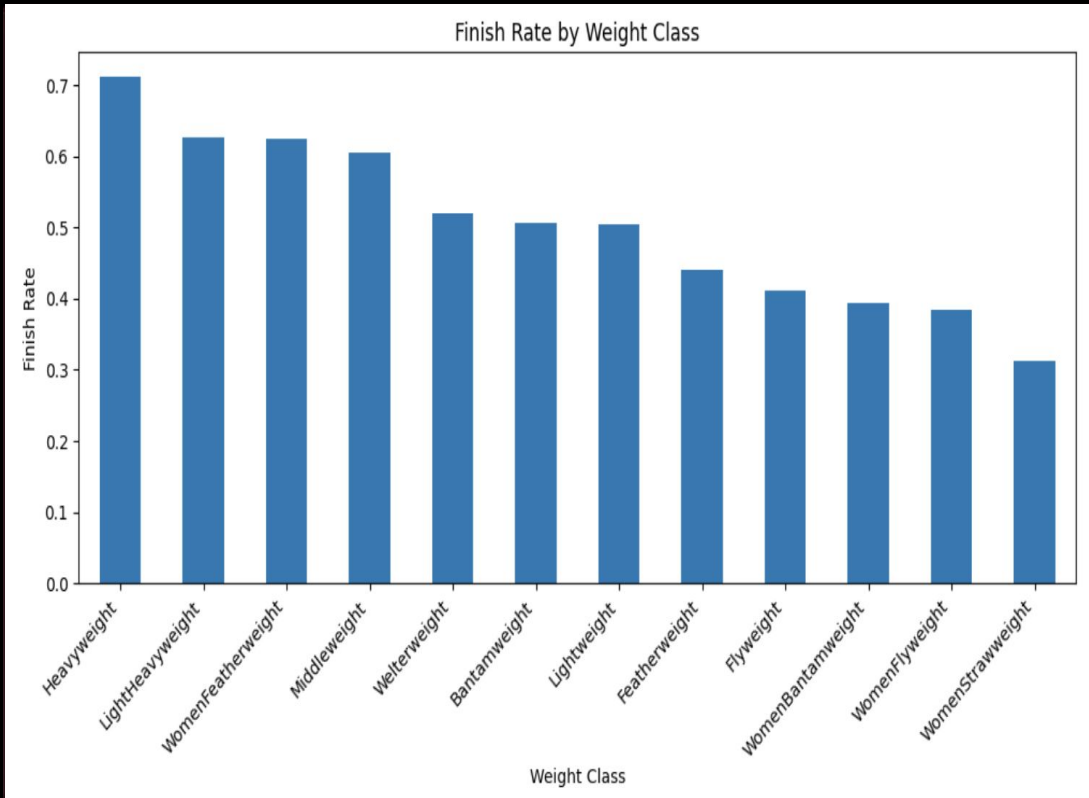
THE DATA ARENA

- **Target:** We defined a 'Finish' as any KO, TKO, Submission, or Doctor Stoppage.
- **Baseline:** The overall dataset is fairly balanced (53% Finishes vs 47% Decisions).
- **The Reality:** Gut instinct isn't enough; the data is messy and requires advanced modeling to find linear separability.



	count	mean	std	min	25%	50%	75%	max
finish_vs_decision								
Decision	2680.0	0.450373	4.997028	-15.0	-3.0	0.0	4.0	17.0
Finish	2997.0	0.188188	5.423883	-17.0	-3.0	0.0	4.0	16.0

THE PHYSICS OF A FINISH (Weight Classes)



WEIGHT CLASS REALITY

- **Highest Finish Rate:** Heavyweight (71%)
- **Lowest Finish Rate:** Women's Strawweight (31%)
- **Action:** We removed 'CatchWeight' and 'OpenWeight' entirely to ensure the model focuses on standardized matchmaking logic.

"Physics matters. If we just bet 'Finish' on every Heavyweight fight, we'd be right 71% of the time. Our model has to be smarter than that baseline to prove its worth."

REMOVING THE FUTURE TO PREDICT IT (Data Prep)

THE THREAT OF DATA LEAKAGE

We removed outcome-related columns that the model wouldn't know before the first bell rings:

- `true_win_by`
- `last_round_time`
- `total_strikes_landed`

Result: 139 clean, pre-fight numeric features to train on.

Removed leakage columns:

```
['Winner', 'true_win_by', 'Referee', 'date', 'date_clean', 'location', 'R_fighter', 'B_fighter']
```

Model dataset shape: (5789, 140)

	title_bout	weight_class	B_avg_KD	B_avg_opp_KD	B_avg_SIG_STR_pct	B_avg_c
0	False	Bantamweight	0.000	0.0	0.420000	
1	False	Middleweight	0.500	0.0	0.660000	
2	False	Heavyweight	NaN	NaN	NaN	
3	False	WomenStrawweight	NaN	NaN	NaN	
4	False	WomenBantamweight	0.125	0.0	0.535625	

5 rows x 140 columns

LET THE **MODELS** FIGHT

(Model Comparison)



PERFORMANCE RESULTS

- **The Contenders:** We tested four distinct architectures, measuring Accuracy, F1, and ROC-AUC simultaneously.
- **The Champion: Gradient Boosting** consistently outperformed the baseline across all three metrics.
- **The Edge:** By achieving the highest Weighted F1-Score (0.568), it offers the best mathematical protection against false predictions.

"We put our algorithms in the cage. Look at the data: Gradient Boosting is our champion. It learns from its own mistakes, round after round, pulling ahead of the pack."

THE **CONFUSION** MATRIX

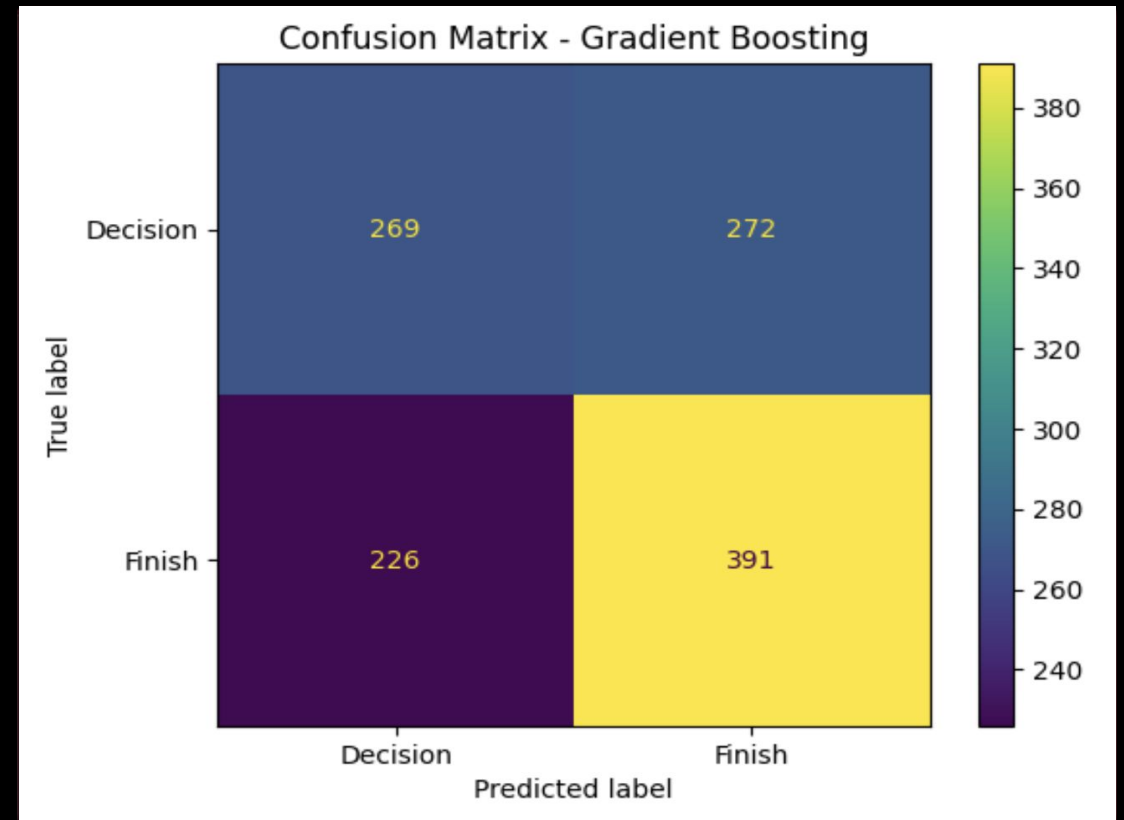
(Evaluation)

UNDERSTANDING THE TRADEOFF

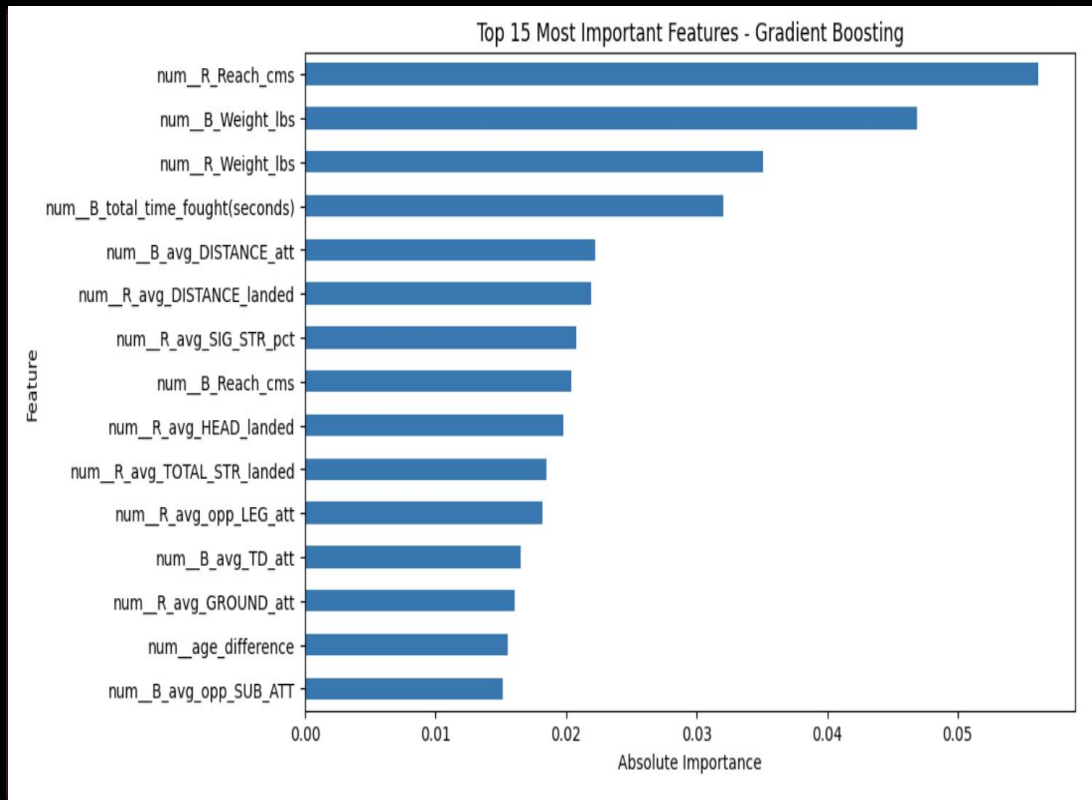
Total Test Set: 1,158 Fights

- **Recall (63%)**: We successfully identified 63% of all actual finishes.
- **Precision (59%)**: When we predict a finish, we are right 59% of the time.

The Reality: Historical stats alone cannot predict a flash knockout, but it does minimize the risk of heavily promoting a "guaranteed knockout" that ends in a slow decision.



THE RECIPE FOR A **FINISH** (Feature Importance)



WHAT THE MACHINE SEES

The top drivers for a Finish aren't just power:

1. **Red Fighter Reach (cm)**
2. **Blue Fighter Weight (lbs)**
3. **Red Fighter Weight (lbs)**
4. **Avg. Distance Attempts**
5. **Age Difference**

The Takeaway: Structural advantages predict finishes better than highlight-reel power.

GENERATING THE **NARRATIVE** (Deployment & LLMs)

FROM CODE TO CONTENT

We built a bridge from our predictive model to a Large Language Model (LLM).

- Model Outputs Probability (e.g., 39%).
- LLM ingests probability and fighter stats.
- LLM outputs automated, marketable fight previews.

```
Model Prediction: - Predicted outcome: Decision - Finish
probability: 0.39 Available fighter statistics: - Red fighter age: 33.0
- Blue fighter age: 22.0 - Red average takedowns landed: 4.0 - Blue
average takedowns landed: 3.25 Important limitation: This prediction is
based only on historical pre-fight data. It does not include injuries,
training camp changes, weight-cut issues, short-notice replacements, or
strategy changes.
```

READY FOR THE OCTAGON

(Verdict & Next Steps)

LIMITATIONS

The data cannot capture injuries, bad weight cuts, or short-notice replacements. This is an **Analytics Support Tool**, not guaranteed betting advice.

DEPLOYMENT PLAN

We recommend building a batch-scoring tool. Analysts can upload an upcoming fight card CSV, and the system will instantly generate finish probabilities and LLM previews for every bout.

Thank You & Q&A

Let's build the future of fight analytics.

The Team: Christian Ian Lopez, Ramon Diehl Verdugo

Technical Review: Full code, EDA, and model pipelines available on GitHub:
github.com/christian-ian-dev/AAI-510-ML-PROJECT